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Research Recruitment
GSA Capital

Re: Quantitative Researcher

To the GSA Capital research team,

My route to quantitative research is unconventional, so I will lead with the work rather than the CV.

This year I designed, built, and ran a systematic trading platform end to end, and used it to test whether a series of candidate FX strategies carried any edge. The infrastructure was production-grade — an autonomous decision agent, deterministic server-side risk gates, live broker integration, streaming market data, and real-time monitoring — but the part I would want to discuss with you is the methodology rather than the plumbing.

I pre-registered acceptance criteria before examining results, held data out strictly for verification, and reported accuracy with Wilson confidence bounds. Four hypotheses went through it: an intraday directional model evaluated across 2,500+ logged forecasts; a five-factor, conviction-gated ensemble ($n = 325$ out-of-sample, 49.8%, lower bound 0.453); time-series carry and trend across eight majors over 14.9 years; and cross-sectional carry across sixteen pairs including the JPY funding crosses. The two backtested families returned Sharpe 0.00 and 0.14 against a standard error of roughly ± 0.26 . A data-mined conditional signal scoring 68.9% in-sample scored 31.3% out-of-sample, with individual cells inverting rather than decaying — the signature of fitting time-correlated noise rather than structure.

Before trusting any of that, I validated the instrument itself. The same engine — lookahead-safe by construction, transaction costs charged on turnover, financing accrued daily, and no fitted parameters — recovered the equity risk premium from index CFDs at Sharpe 0.55 net of costs. The nulls were measurements rather than malfunctions. I had earlier validated the engine against synthetic series with known properties: trending, random-walk, and drawdown regimes.

I am aware that a negative result is an unusual thing to lead a letter with. I do it deliberately. The discipline required to retire four of one's own hypotheses, and to build the positive control that makes retiring them credible, strikes me as the substance of the work rather than an aside to it — and as the difference between a robust trading algorithm and an overfitted one.

My formal credentials are a B.S. in Finance (Auburn, magna cum laude) and dual master's degrees in Applied Economics and Applied Analytics at Boston College (3.87 GPA, expected 2027), with graduate coursework in econometrics, machine learning, and statistical modelling. My professional background is twenty-three years as a U.S. Army officer, currently an information operations planner. That is not quantitative finance. It is two decades of forming and defending conclusions under genuine uncertainty, in front of audiences who ask hard questions and are unmoved by confident presentation.

[Availability: add one line here on your timing, retirement date, and willingness to relocate to London — GSA will want to know.]

I recognise this is not the profile you typically see, and I would not ask you to take the methodology on faith: the platform, the code, the incident write-ups, and every result above are public and reproducible from free data with a single command. If the work suggests someone worth half an hour, I would welcome the conversation.

Respectfully,

David N. Belyn